

CrewAI Comparison Report

Sequential vs. Hierarchical vs. Single-Agent

1. Introduction

The objective of this experiment was to compare three different AI agent architectures for solving the business problem of selecting the best laptop for a "**Budget Laptops for Students**" marketing campaign. The three approaches evaluated were:

- **Single-Agent (LangGraph)** – One intelligent agent performs the complete workflow from product analysis to the final marketing recommendation.
- **CrewAI Sequential Workflow** – Three specialized agents collaborate in a predefined sequence where each agent performs a dedicated task before passing its output to the next agent.
- **CrewAI Hierarchical Workflow** – A Campaign Manager coordinates multiple specialist agents by delegating tasks, reviewing their outputs, and assembling the final marketing brief.

The comparison focuses on workflow organization, token usage, execution cost, latency, output quality, reliability, and suitability for the given business problem.

2. Experimental Setup

The business objective was to recommend the best laptop for a student-focused marketing campaign. Three specialist agents were designed with clearly separated responsibilities.

Data Analyst

Responsible for analyzing the internal product catalog, comparing laptop prices and stock availability, and calculating the price difference between the best candidates.

Market Researcher

Responsible for researching competitor pricing and identifying which shortlisted laptop has the strongest competitive advantage in the market.

Marketing Strategist

Responsible for preparing a concise stakeholder-ready marketing brief using only verified information received from the previous agents.

For the hierarchical implementation, a fourth agent—the **Campaign Manager**—was introduced. Rather than performing the work directly, the manager delegated tasks to the appropriate specialist, reviewed their outputs, and ensured consistency before producing the final result.

3. Workflow Comparison

Single-Agent LangGraph

The LangGraph solution uses one intelligent agent that completes the entire workflow within a single reasoning process. The agent plans its approach, retrieves the required information, performs the necessary reasoning, and generates the final marketing recommendation.

Advantages

- Lowest computational overhead
- Fastest execution
- Lowest token consumption
- Simple implementation
- Easy debugging
- Minimal coordination required

Disadvantages

- Performs every responsibility itself
- No separation of expertise
- Greater chance of mixing business analysis with marketing reasoning
- Difficult to trace which reasoning step produced a specific conclusion

CrewAI Sequential Workflow

The sequential workflow divides the problem among three specialized agents. Each agent performs a single responsibility before passing structured information to the next stage.

Workflow:

Data Analyst



Market Researcher



Marketing Strategist

The Data Analyst first identifies suitable laptops from the internal catalog. The Market Researcher then compares competitor prices using the shortlisted candidates. Finally, the Marketing Strategist

prepares the final stakeholder report using only verified information received from the previous agents.

Advantages

- Clear separation of responsibilities
- Better explainability
- Easier to verify intermediate outputs
- Lower risk of hallucinated numerical values
- Better maintainability
- Easier debugging because each stage can be tested independently

Disadvantages

- More token usage
- Higher execution cost
- Additional latency because each agent waits for the previous one to finish
- Context passing introduces additional prompt overhead

CrewAI Hierarchical Workflow

The hierarchical workflow introduces a Campaign Manager who coordinates the work performed by specialist agents.

Workflow:

Campaign Manager



Data Analyst



Market Researcher



Marketing Strategist



Campaign Manager Review



Final Output

Rather than manually defining the workflow, the Campaign Manager decides which agent should perform each subtask, reviews their outputs, and assembles the final recommendation.

Advantages

- Dynamic task delegation
- Additional verification stage
- Better suited for complex projects
- Easily scalable
- Flexible when workflow order is unknown

Disadvantages

- Highest token usage
- Highest execution cost
- Longest execution time
- Increased implementation complexity
- More coordination failures possible
- Additional manager prompts increase overhead

4. Cost and Performance Comparison

Feature	Single-Agent	Sequential CrewAI	Hierarchical CrewAI
Token Usage	Lowest	Moderate	Highest
Execution Cost	Lowest	Moderate	Highest
Latency	Fastest	Moderate	Slowest
Number of Agents	1	3	4
Prompt Overhead	Lowest	Moderate	Highest
Context Passing	None	Moderate	High
Task Delegation	None	Manual	Automatic

Workflow Complexity	Low	Medium	High
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5. Output Quality Comparison

Single-Agent

The single-agent implementation produced a complete recommendation quickly and efficiently. However, because the same model handled product analysis, market research, and marketing writing simultaneously, there was a greater possibility of blending internal and competitor information without clear traceability.

Sequential CrewAI

The sequential implementation generated the highest-quality output for this assignment. Each agent specialized in one responsibility, making the workflow more organized and reducing the risk of factual errors. Since every stage received structured input from the previous stage, the final marketing brief remained consistent and easy to verify.

Hierarchical CrewAI

The hierarchical workflow produced output comparable in quality to the sequential workflow but required additional planning and delegation. The Campaign Manager improved coordination by reviewing specialist outputs before producing the final recommendation. However, for a fixed three-step workflow, the additional management overhead provided only a small improvement in quality while increasing execution time and computational cost.

6. Success Criteria Evaluation

The following criteria were used to evaluate the quality of the generated marketing brief.

1. Factual Grounding

Every numerical value included in the final report must originate from verified tool outputs rather than being generated by the language model.

Evaluation: Successfully achieved.

2. Completeness

The final report should contain:

- One recommended laptop
- Internal product information
- Competitor pricing
- Business justification
- Marketing recommendation

Evaluation: Successfully achieved.

3. Tone and Audience

The report should:

- Remain under 200 words
- Be understandable to non-technical stakeholders
- Maintain a professional business tone

Evaluation: Successfully achieved.

7. Overall Discussion

The experiment demonstrated that increasing the number of specialized agents improves task organization and separation of responsibilities but also increases computational overhead. While the single-agent solution is efficient and inexpensive, it performs all reasoning within one context, making it more difficult to isolate responsibilities or validate intermediate reasoning.

The sequential CrewAI workflow offers a balanced approach by assigning independent responsibilities to specialist agents while maintaining a predictable execution order. This design improves transparency, traceability, and factual grounding without introducing unnecessary management complexity.

The hierarchical workflow demonstrates CrewAI's ability to coordinate multiple specialists dynamically through a manager agent. Although this architecture is highly flexible, its benefits become significant only for larger, less structured problems where task ordering cannot be predetermined.

8. Final Conclusion

For the "**Budget Laptops for Students**" campaign, the **Sequential CrewAI workflow** proved to be the most appropriate solution. The business problem naturally decomposed into three clearly defined stages—product analysis, competitor research, and marketing strategy—which aligned perfectly with the capabilities of specialized agents. Compared with the single-agent LangGraph

implementation, the sequential workflow improved traceability, modularity, and factual consistency while maintaining reasonable execution cost and complexity. Although the hierarchical workflow demonstrated advanced multi-agent collaboration and delegation, its additional planning overhead did not produce sufficient quality improvements for this relatively simple, predefined workflow. Therefore, the sequential CrewAI architecture offers the best balance between output quality, computational efficiency, maintainability, and overall system design for this assignment.